PROJECT PRESENTATION

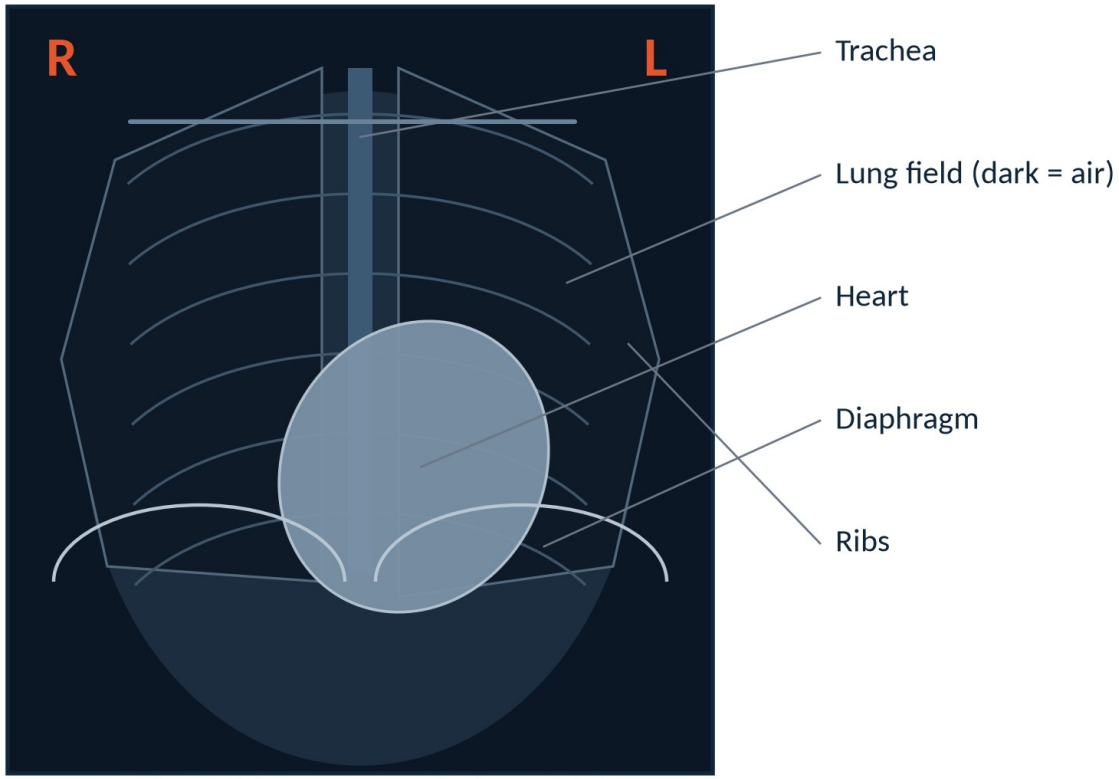
Medical Visual Question Answering for Chest X-rays

A knowledge-grounded approach with CheXagent-2-3b: literature review, baseline tests and retrieval-augmented design

Supervisor: [Supervisor name, designation]
Team members: [Name 1, roll no.]
[Name 2, roll no.]
[Name 3, roll no.]

[Institution name]
[Department / programme]
[Month Year]

A chest X-ray is a shadow picture: dark means air, white means dense tissue



patient's right appears on the image's left

What a radiologist looks for

Cardiomegaly
heart wider than half the chest (cardiothoracic ratio > 0.5)

Pleural effusion
fluid blunts the sharp corner at the lung base

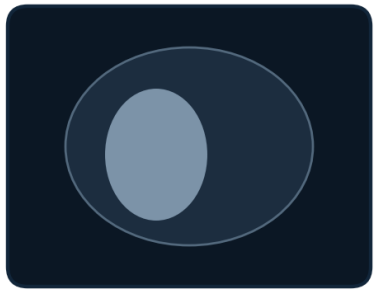
Pneumothorax
air outside the lung: dark area with no lung markings

Atelectasis
collapsed lung looks whiter; structures pull toward it

Four findings used throughout this project: heart size, fluid, air outside the lung, and collapse.

Sources: radiology teaching material (cardiothoracic ratio, costophrenic angle, pleural line); schematic drawn for this report, not a real image.

Medical VQA: show the model an X-ray and a question, then check its free-text answer



chest X-ray

Is the heart enlarged?

→ **Vision-language model (CheXagent-2-3b)** →

free-text answer

scored against the dataset's gold answer

Four kinds of question, two kinds of difficulty

Yes / no *perception*	**Which side** *perception*	**How severe** *perception*	**Knowledge** *medical knowledge*
Is there a pleural effusion?	Which side is the finding on?	Is the heart enlarged?	What causes this disease?

Two kinds of difficulty: seeing the finding (perception) and knowing about the disease (knowledge).

Datasets: SLAKE and VQA-RAD, chest X-ray questions only. Model: StanfordAIMI/CheXagent-2-3b.

Of six papers reviewed, three shaped the project; none combines retrieval with chest X-rays

	Chest X-ray focus	Writes free text	Uses retrieval	Clinician / blind rating	Textbook source	what we took from it
CheXagent (2024)	●	●	○	●	○	our baseline model; weakest on left/right and severity (77-79%)
RULE (2024)	○	●	●	○	○	naive retrieval hurts: 47-59% of its errors follow the text
Med-Flamingo (2023)	○	●	○	●	○	automated scores disagree with clinicians: rate answers blind
LLaVA-Med (2023)	○	●	○	○	○	cheap curriculum tuning; GPT-4-written training data
PMC-VQA (2023)	○	●	○	○	○	caption-made questions can be answered from text alone
PubMedCLIP (2023)	○	○	○	○	○	the image encoder matters more than the text encoder
This project	●	●	●	○	●	**none of the papers combines all of these (hollow = planned)**

CheXagent is our baseline, RULE warns about retrieval, Med-Flamingo says to rate answers by hand.

Chen et al. 2024 (CheXagent); Xia et al., EMNLP 2024 (RULE); Moor et al. 2023 (Med-Flamingo); Eslami et al., EACL 2023 (PubMedCLIP); Li et al. 2023 (LLaVA-Med); Zhang et al. 2023 (PMC-VQA).

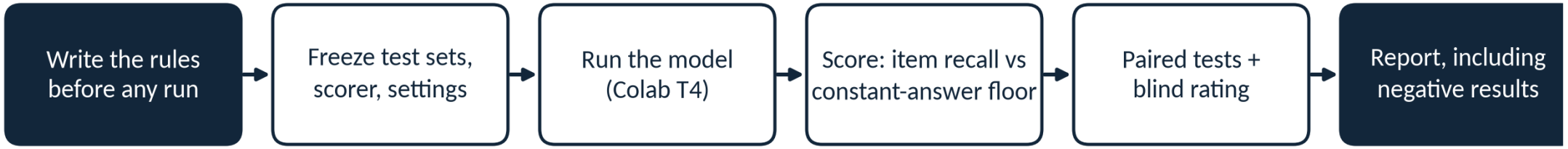
Each finding in the literature became a concrete design decision

What the papers show	What we decided to do
CheXagent is weakest on left/right and severity	Test perception on its own: mirror-image test, heart-size wordings
Naive retrieval makes answers worse (RULE)	Few passages, a no-harm check, and an arm that gives the disease name without any text
Automatic scores disagree with clinicians	Blind 0-1-2 rating, plus a constant-answer floor that a real system must beat
No reviewed paper retrieves textbook knowledge for chest X-rays	BM25 search over MedlinePlus + medical textbooks, independent of the test data

Research question: can retrieved textbook knowledge make a chest X-ray model answer knowledge questions better?

Findings from CheXagent (77-79% on fine-grained reasoning), RULE (naive retrieval over-trust) and Med-Flamingo (clinician rating).

Rules are written before any run, and negative results are reported



Test sets used so far (all chest X-rays, SLAKE + VQA-RAD)

219	130	140	120	47	130
baseline questions	knowledge questions	left / right questions	mirror-image pairs	heart-size questions	images with a disease label

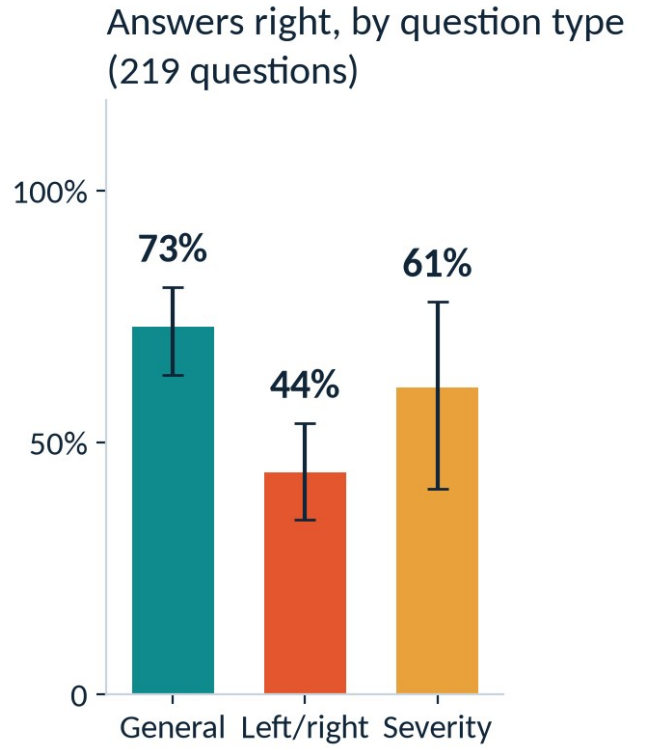
Statistics: bootstrap resampled by image (several questions share an image), exact McNemar test, exact sign test.

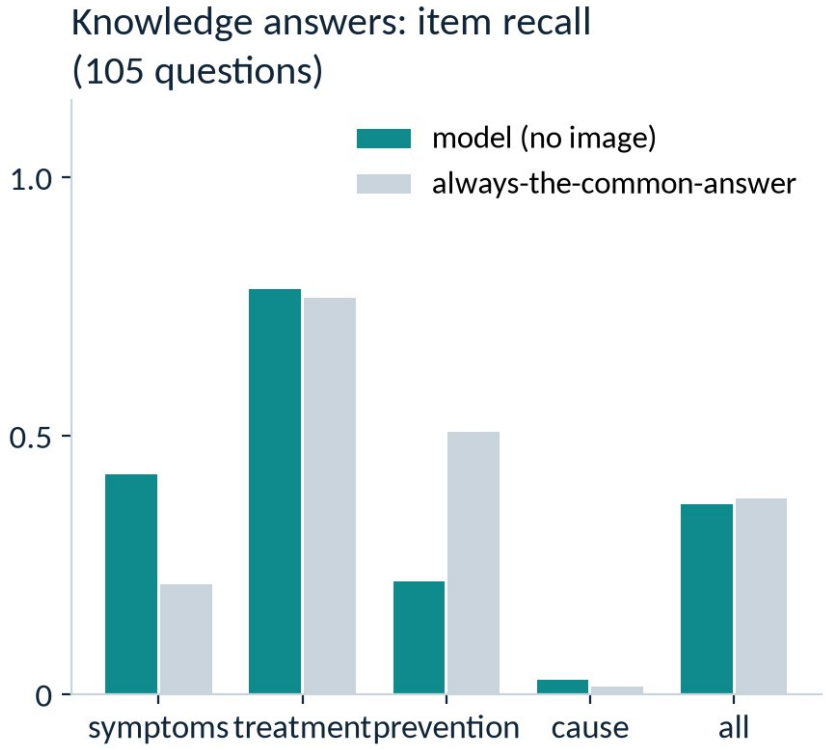
Scorer: item recall (share of the gold answer's items the answer covers); frozen as version 2.

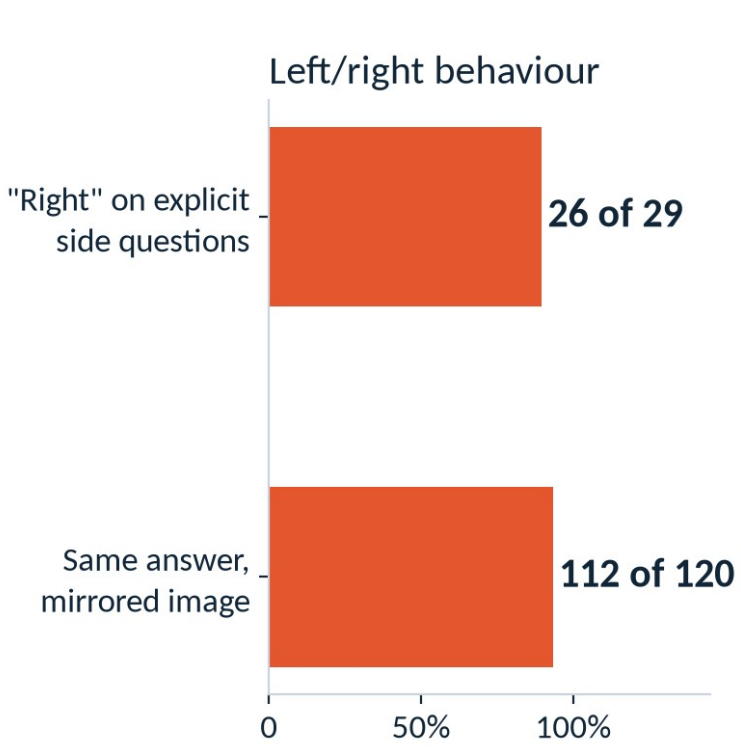
Pre-registration: test sets, scorer and success conditions were frozen first, then amended in writing when needed.

Documented in docs/evaluation_protocol.md (6 amendments) and docs/phase2_rag_plan.md.

Our baseline tests: 58% overall, but the left/right and knowledge scores do not hold up





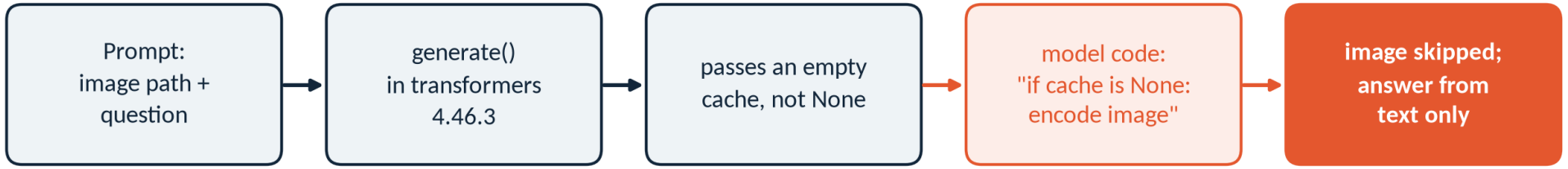


Left/right: answers on a mirrored image were unchanged in 112 of 120 pairs. Knowledge: no better than a fixed answer.

Runs 2-4 (219 baseline questions, 105 knowledge questions, 120 mirror-image pairs). Error bars: 95% Wilson intervals.

Turning point: the image never reached the model, so earlier results are a text-only control

What went wrong



How we noticed

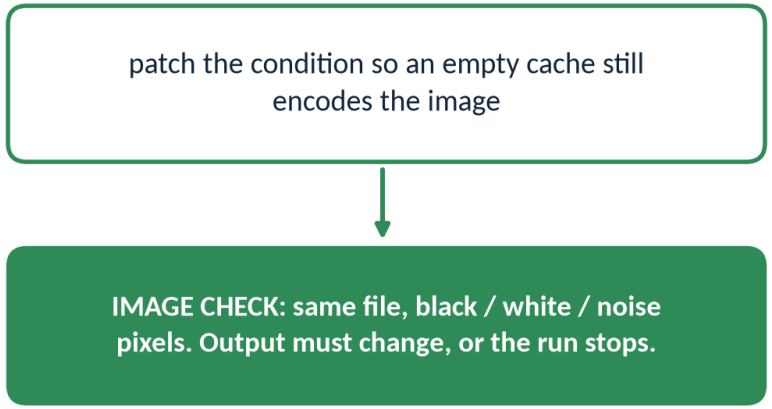
130 of 130
images got the same answer ('Pneumonia' / 'No Finding')

0.002-0.009
spread of yes/no scores across 130 different X-rays

Earlier results are kept as a text-only control.

Every notebook now carries the check.

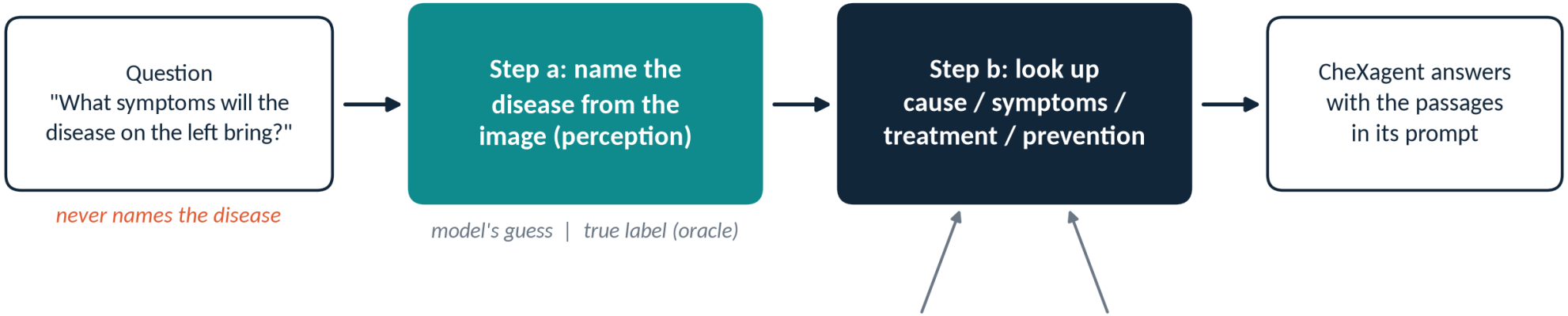
The fix



What stays valid: how much of these benchmarks a text-only model solves. What is withdrawn: every conclusion about perception.

transformers 4.46.3 passes an empty cache; CheXagent's code only encodes the image when the cache is None. Recorded in experiment_log.md, entry 17.

Phase 2 design: first name the disease from the image, then retrieve what is known about it



Why two steps: SLAKE's answers are a lookup by disease

same disease -> same gold answer, on every image:

Pneumonia cause: bacterial infection

Pneumonia treatment: medical therapy, supportive therapy

Cardiomegaly cause: hypertension, dilated cardiomyopathy

8 chest diseases cover all 130 labelled images

Why: SLAKE's knowledge answers are a lookup keyed by the disease, and the question never names it.

Corpora: MedlinePlus (US National Library of Medicine, public domain) and medical textbooks (MedRAG), searched with our own BM25.

Six arms separate what retrieval adds from what knowing the disease adds

Arm	Disease named	Retrieved text	Role
A1	none	none	fair baseline (same answer style)
A2	none	from the question alone	oracle subset: shows why disease matters
A3	true disease (oracle)	none	upper bound: value of knowing the disease
A4	true disease (oracle)	yes	upper bound: value of retrieval
A5	model's own guess	none	real: guess alone
A6	model's own guess	yes	the system

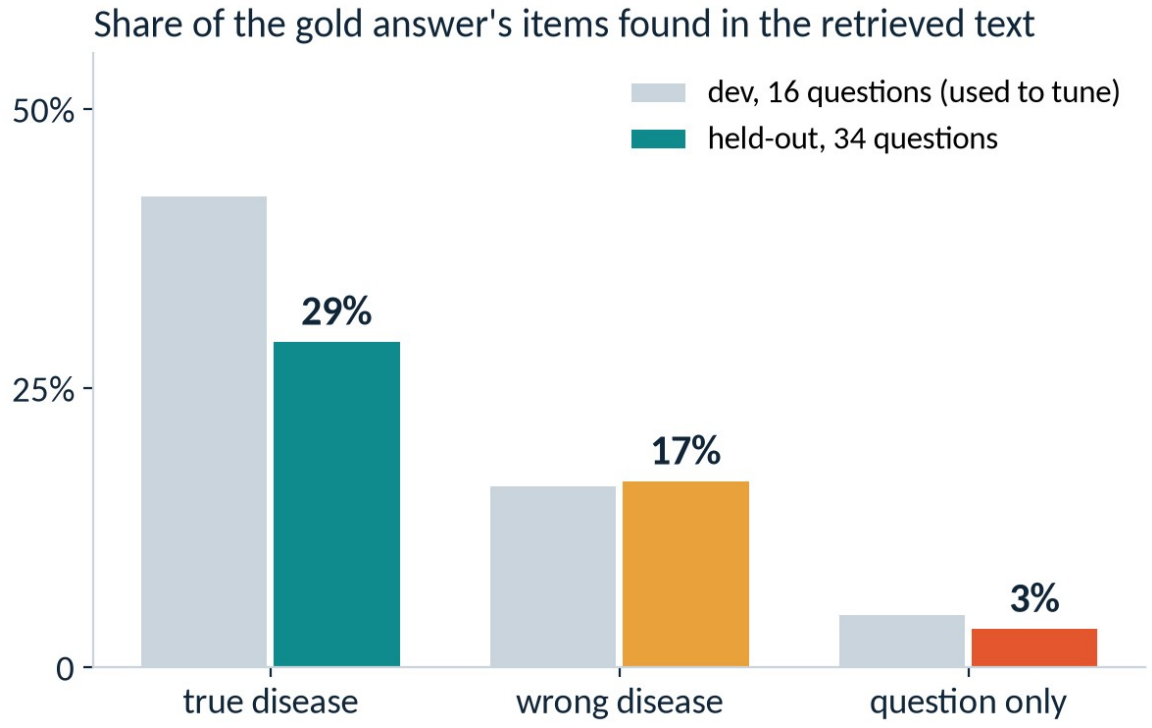
oracle arms use SLAKE's own label: upper bounds, never reported as the result

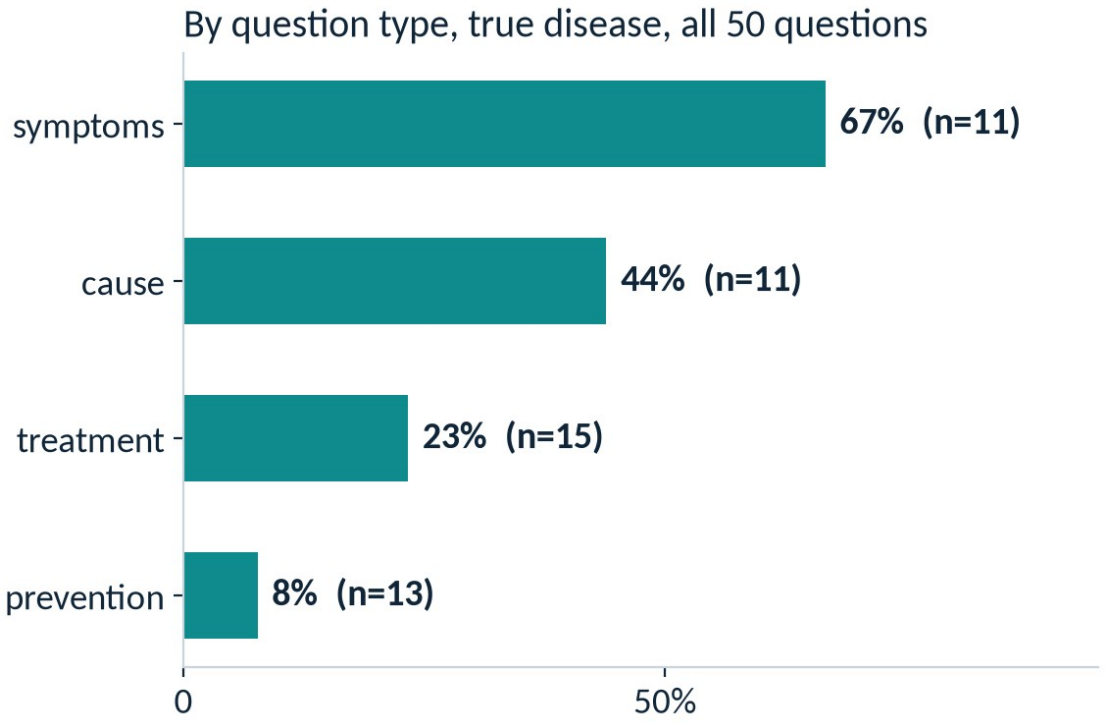
real arms use only what the model outputs

Success needs more than a score gain: the system (A6) must beat both the fair baseline (A1) and the guess alone (A5).

Defined in docs/phase2_rag_plan.md and Amendment 5 of the evaluation protocol. Oracle arms are upper bounds only.

Retrieval check, run without any model: the corpus holds the answer only when the disease is right

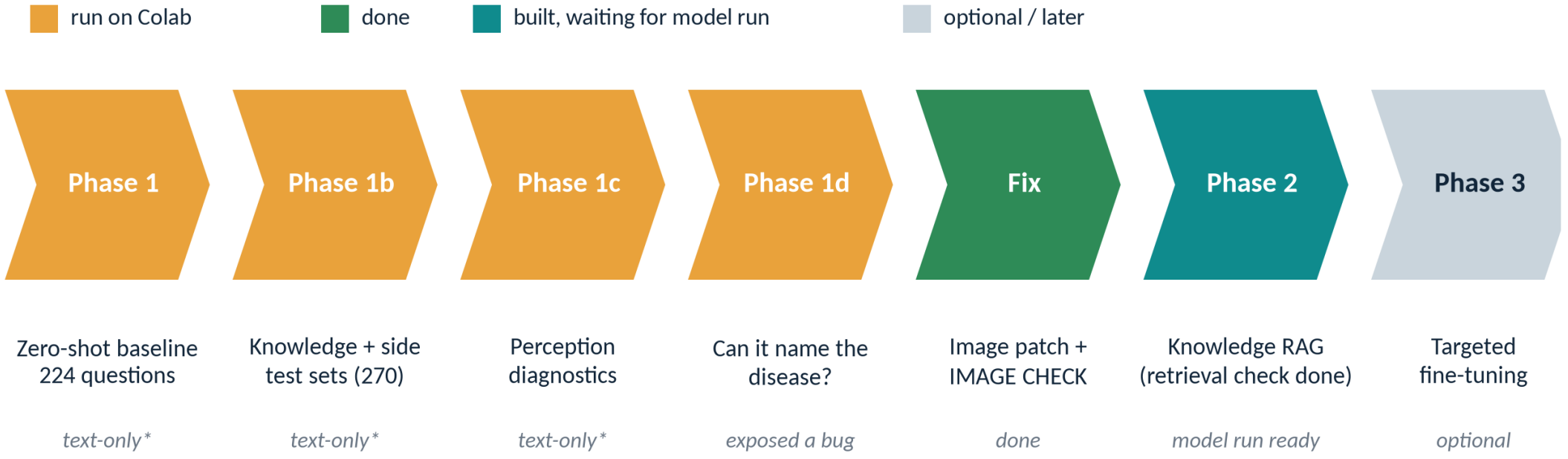




Held-out: 29% of gold items found with the true disease, 17% with a wrong disease, 3% from the question alone.

Strict coverage = gold item words inside one sentence of the retrieved text. 50 oracle-evaluable questions; setting frozen on 16 dev questions only.

Where we stand: pipeline and design are ready, the image-visible model runs come next



* the model could not see the image in these runs; the results are a text-only control (see the turning point)

Next: rerun notebooks 04 and 05 (then 01-03) with the image fix; then blind rating. No clinical claims are made.

Limits: zero-shot only, small test sets, one model, SLAKE answers are templates, and no clinician review yet.